

DoCircuits Circuit Simulator

In this month's software focus we take a closer look at an innovative online circuit simulator named 'DoCircuits.' An offering of Sparsha Technologies, this circuit simulator allows you to store your circuits on the cloud, making it available for use and analysis anywhere on earth

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Sparsha is a learning technology startup headquartered at STEP, Indian Institute of Technology, Kharagpur with an extended R&D arm in Bengaluru. Sparsha works with education content providers, online learning companies, publishers, universities and autonomous institutes to develop custom solutions for activity-based learning solutions across K-12 and higher education domains. Let's dive in straight into the functionalities of DoCircuits and how it differs from an offline simulator.

DoCircuits

An electronic circuit simulator is a software that uses mathematical models to replicate the behaviour of an actual electronic device or circuit. Simulation software lets you model a circuit operation and is therefore an invaluable analysis tool. Its highly accurate modelling capability is one of the main reasons why colleges and universities use this type of software for teaching electronics and electrical engineering classes. A simulation software engages the users by integrating them into the learning experience. These kinds of interactions engage learners to analyse, organise, synthesise and evaluate content.

There are several issues with the solutions offered presently in the simulation domain. The software needs to be downloaded/installed. The software has a steep learning curve requiring help of an expert to guide you side-by-side.

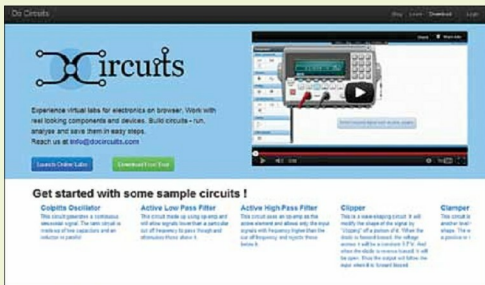


Fig. 1: DoCircuits - Home



Fig. 2: DoCircuits - online simulator

User-experience isn't exactly a priority with these software. One of the bigger issues is that people who are just starting out in this field are in no position to use the high-end simulators. They need to get an in-depth knowledge before beginning work.

The biggest obstacle is that most of today's simulators are offline versions;

you install them in your machine and then work from there. If you want to continue work on the model at a place other than your home machine, you need to carry the file with you, have the simulator installed there and continue, which is, frankly, a very arduous process.

This way, DoCircuits is a boon for simulation students, hobbyists and